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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	T010NG
Project / Solution Name	Kotak
Name (Leader)	Hee Zi Jie
Email & Phone (Leader)	heezijie2004@gmail.com, +60125118986
University Affiliation(s)	Asia Pacific University of Technology & Innovation (APU), University of Reading and University of Sheffield
Live Prototype URL	https://kotak-alaskxntxna.replit.app (mirror: https://kotak.tp073654.workers.dev)
Code Repository URL (public / access granted to judges)	https://github.com/TolongLabs/sdc-hack

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

A user opens Kotak in the browser. The React app loads dated JSON snapshots built from O*NET, the Anthropic Economic Index and DOSM Job Market Insights, and computes every verdict, fit score and roadmap in the browser with fixed rules. Answers stay in the browser's localStorage.

The app calls a Hono API on the same origin. The API matches typed job titles and writes cited explanations with Claude Haiku 5.5, then checks each explanation against the rules and falls back to a template if it fails. It sends Kopi's lines to ElevenLabs for speech, and caches both by a hash of the request. Ask Kopi and the MCP endpoint call 6 read-only tools over the same rules; one tool reads anonymous check-up totals from Neon Postgres.

Before the demo, a RunPod serverless GPU ran Qwen3.8-27B to build each job's 6 skills and 18 questions, which ship inside the snapshots.

GitHub Actions builds and deploys the app and API to Cloudflare Workers on every merge to main, and refreshes the data every Saturday. Replit publishes the same code from the team's account. Keys live only in GitHub, Cloudflare and Replit secrets.

System block diagram: see Appendix A.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 7.0	The language of the whole codebase	Apache 2.0
Bun 1.4	Runtime, package manager, test runner and Replit server	MIT
React 19.3	The web app's user interface	MIT
Vite 8.3	Builds and serves the web app	MIT

Name & Version	Purpose in Project	License
Tailwind CSS 4.3	Styling and design tokens	MIT
Recharts 3.10	The Future Skills Radar and the roadmap charts	MIT
Lucide React 1.61	Icons	ISC
pixi.js 6.5 and pixi-live2d-display 0.4	Renders Kopi, the Live2D cat	MIT
Hono 4.13	The API server on Cloudflare Workers and Bun	MIT
Zod 4.6	Validates every request, response and data file	MIT
Anthropic TypeScript SDK 1.1	Calls the Claude API	Apache 2.0
MCP server SDK 2.3	Serves Kotak's tools to AI assistants at /api/mcp	Apache 2.0
Neon serverless driver 1.2	Connects the API to Postgres	MIT
Wrangler 4.151	Deploys to Cloudflare Workers	MIT or Apache 2.0
Playwright Core 1.64 and Clearcote 0.60	Drives the browser for the JobStreet research scraper	Apache 2.0, BSD-3
Live2D Cubism Core for Web	Runtime that plays Live2D models, loaded from Live2D's CDN	Live2D Proprietary Software License
Atkinson Hyperlegible and Shantell Sans fonts	Body, mono and Kopi's handwritten type, from Google Fonts	SIL Open Font License 1.1

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Claude API, Anthropic	Matches job titles, explains verdicts and runs Ask Kopi	Paid
Text to Speech API, ElevenLabs	Kopi's voice	Paid
RunPod Serverless API, RunPod	The one-time GPU batch that built each job's skills and questions	Sponsor-provided credit
Google Fonts, Google	Serves the app's fonts	Free
Live2D Cubism CDN, Live2D	Serves the Cubism Core runtime for Kopi	Free

The data sources in section 4.2 are downloaded files, not live APIs: the app never calls them at runtime. API keys are not shown in this document.

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Deployments, Replit	Publishes the live demo from main on the team's own account	Replit Core (sponsor-provided)
Cloudflare Workers, Cloudflare	The mirror deployment: app, API, KV cache and a Durable Object for MCP and chat	Workers Free; global, Durable Object in APAC
Neon Postgres, Neon	Anonymous check-up totals, research job listings and admin accounts	Free; aws-ap-southeast-1, Postgres 18
RunPod Serverless, RunPod	The vLLM GPU endpoint for the one-time batch; 0 workers when idle	Sponsor credit; one H100 80 GB GPU
GitHub and GitHub Actions, GitHub	Code hosting, CI, deploys on every merge to main and the Saturday data refresh	Free

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Claude Haiku 5.5 (claude-haiku-5-5)	Anthropic, Claude API	Pre-trained	Anthropic Commercial Terms of Service
ElevenLabs eleven_v4 and eleven_v4_turbo, voice Johari	ElevenLabs	Pre-trained	ElevenLabs Terms of Service
Qwen3.8-27B-FP8	Qwen, run on RunPod with vLLM	Pre-trained	Apache 2.0

No model is fine-tuned or self-trained.

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
O*NET 30.1 Database	https://www.onetcenter.org/database.html	Public	CC BY 4.0	N
Anthropic Economic Index, April and May 2026	https://huggingface.co/datasets/Anthropic/EconomicIndex	Public	Data CC BY; code MIT	N
DOSM Job Market Insights, Q3 2019 to Q2 2026	https://www.dosm.gov.my	Public	DOSM terms, with source credit	N
OpenDOSM skill-related underemployment by age	https://open.dosm.gov.my/data-catalogue/lfs_qtr_sru_age	Public	CC BY 4.0; one figure cited	N
UN SDG indicator 4.4.1, Malaysia	https://unstats.un.org/sdgapi/	Public	UN terms of use; one baseline cited	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Kotak skills and questions for 923 jobs	Built from O*NET by the RunPod batch	Synthetic, from public data	The team's own work	N
JobStreet job listings	Collected by the team's research scraper from https://my.jobstreet.com	Self-collected	JobStreet's terms apply	N; job ads only
Demo check-up profiles	Written by the team	Synthetic	The team's own work	N

MyCOL and Epoch AI are cited as context only; Kotak loads no data from them. The research listings feed no verdict, score or roadmap.

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code, with Claude models	Code, reviews, research, documentation and pull requests	Major
Codex CLI, OpenCode and Antigravity CLI, with GPT, DeepSeek, GLM and Gemini models	Parallel code changes from written briefs	Moderate
Claude, GPT and Gemini chat models	The field simulation and design review in docs/research	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every change went through a pull request that ran type-checks and a build in CI before a member merged it. The team checked each feature by running the app in a browser at phone and desktop widths, not by trusting the tool's report. Facts in the app and in these documents link a dated public source. AI-drafted skills carry a visible label in the app until a person reviews them. The team owns every merged line and corrected tool mistakes before merging.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Platform, Core	Hosts the live demo, published from main on the team's own account
RunPod	Credits	A serverless H100 ran the batch: 923 jobs, 783 seconds for the full run, about US$1.25
SDC mentors (KrackedDevs)	Mentoring	One mentor session on 23 October 2026; feedback recorded in #98

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

RunPod was essential to build Kotak's data: it turned O*NET's tasks into 6 skills and 18 questions for each of 923 jobs, which no person could write by the deadline. The batch runs once, so the live app does not depend on RunPod, and the Claude Message Batches API could replace it. Replit hosts the submitted live demo; the same build also runs on Cloudflare Workers, so the hosting is replaceable.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Kotak has no user accounts. A check-up runs in guest mode: the job, goal and answers stay in the browser's localStorage, and Settings > Reset All clears them.

The app sends a typed job title to the Claude API for matching, and a role's skill data for explanations. Ask Kopi sends the user's message, and the results of tools run on their check-up, to Claude. Kopi's lines go to ElevenLabs for speech. Neither request carries a name, email or ID. The API caches answers and audio by a hash of the request.

Postgres stores anonymous check-ups as a role code, 6 skill levels, a sample flag and a date, and shows totals only for groups of 10 or more. The research scraper stores public job ads, not personal profiles. Keys live only in host secrets.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

- **Hallucination**: Claude only explains verdicts that fixed rules set. A check rejects any explanation that changes a verdict or lacks a dated citation, and a template replaces it.
- **Wrong job match**: the user confirms every AI title match.
- **Unreviewed skills**: an open model drafted each job's skills. A validator rejects tasks from other jobs, and the app labels the skills AI-drafted.
- **Thin trend data**: the Economic Index has two monthly points, so 12-month projections are rough. Every target is labelled an estimate, and thin data shows Unknown.
- **Data bias**: O*NET describes US jobs, and DOSM counts only online job ads.
- **Single providers and rate limits**: cached answers, templates and text cover Claude or ElevenLabs outages and HTTP 429 or 529 errors. A limit of 30 Claude calls a minute protects the key.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

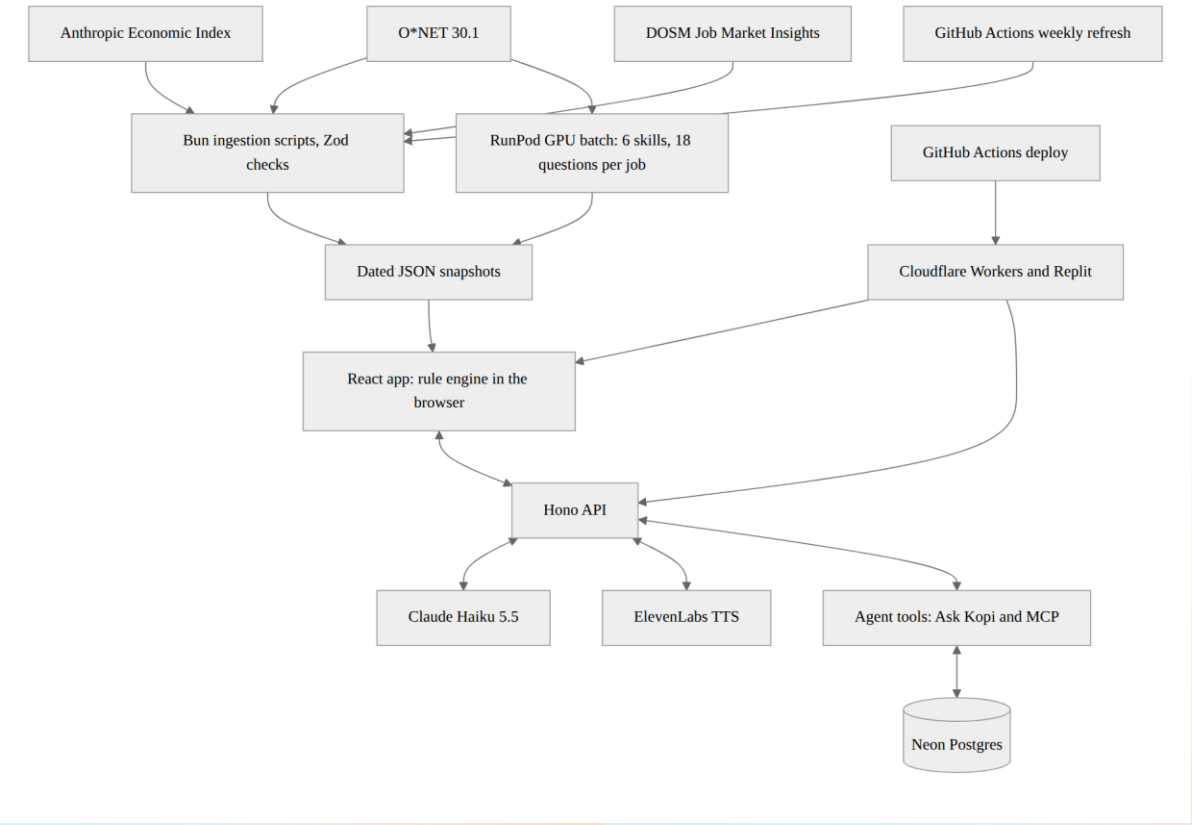
8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



24 October 2026

Appendix A: System Block Diagram



Data flows top to bottom: dated public data and the RunPod batch build the snapshots, the browser computes every verdict, and the API adds Claude, ElevenLabs and the agent tools.

Appendix B: Replit Proof of Use

Handbook 10.6 asks for proof that Replit was used, showing the team's own account and project. The following pages show the live deployment at https://kotak-alaskxntxna.replit.app and the project https://replit.com/@alaskxntxna/Kotak with its publish and Git history, from the team leader's account. The same file is submission/REPLIT PROOF OF USE.pdf in the repository.

Replit Proof of Use

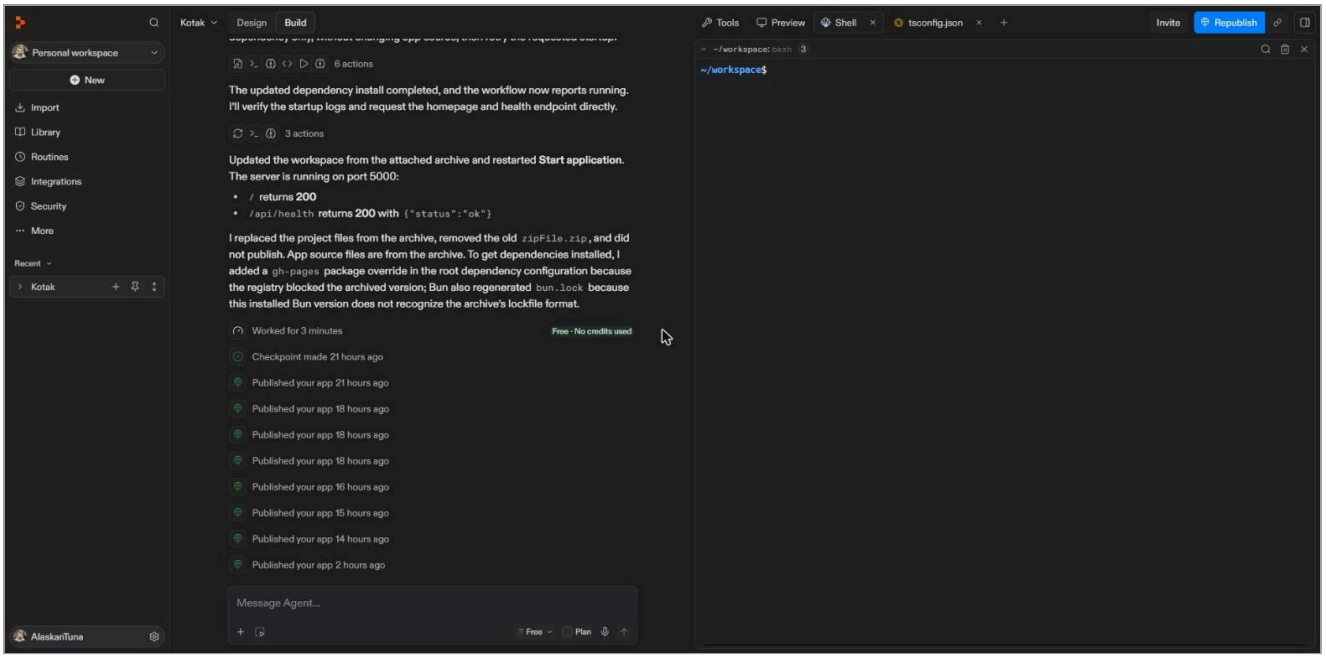
Kotak · Team T010NG · SDC Hackathon 2026, Domain 1 · Captured 23 October 2026

How Kotak Uses Replit

Replit account	alaskxntxna (display name AlaskanTuna), the team leader's own account
Project	https://replit.com/@alaskxntxna/Kotak
Live deployment	https://kotak-alaskxntxna.replit.app, status Live, sharing Public
Deployment type	Replit Autoscale, region Asia, 2 vCPU / 4 GiB RAM
Source	GitHub repository TolongLabs/sdc-hack, connected through Replit Git (branch `github-main`)
Run configuration	`.replit`: `bun install`, `bun run build`, then `bun run start` on port 5000
Secrets	ANTHROPIC_API_KEY, ELEVENLABS_API_KEY and SESSION_SECRET in Replit Secrets; values stay hidden
Publish history	9 publishes by AlaskanTuna between 22 and 23 October 2026

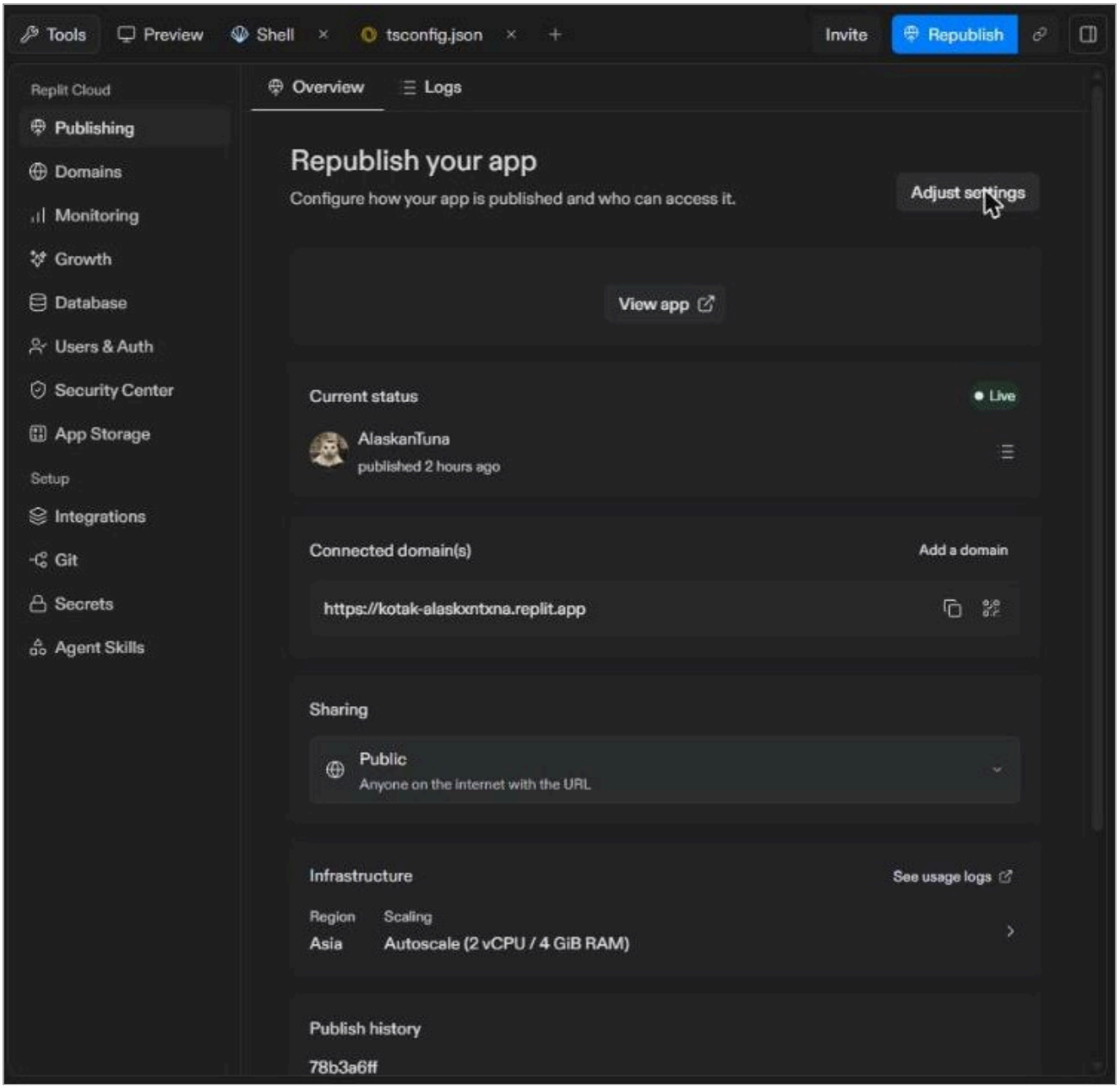
Handbook 10.6 accepts a live deployment URL or a project link with its history as proof. This document shows both, from the team's own account. The figures follow.

Figure 1



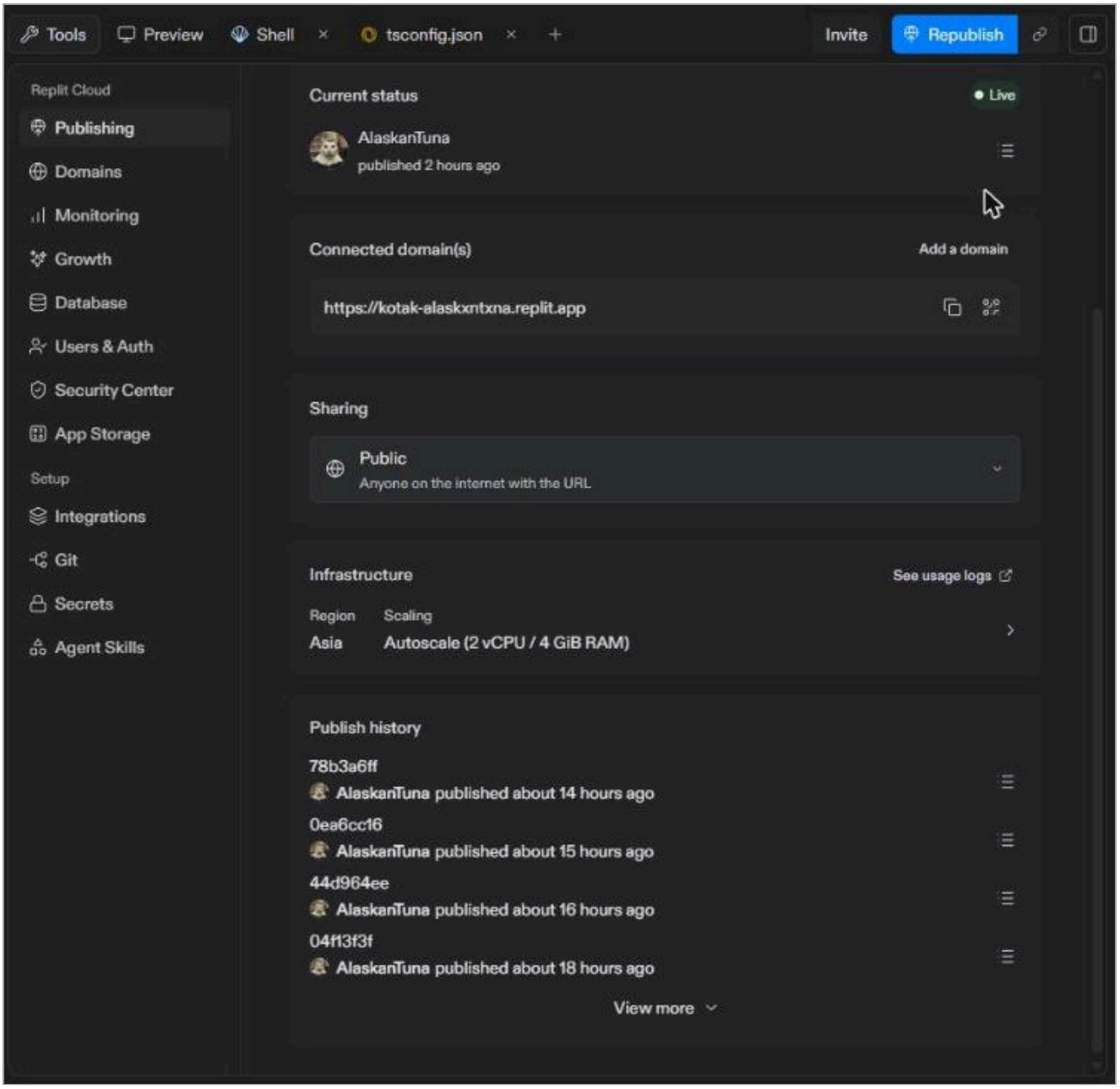
The Kotak project in the AlaskanTuna workspace, with its publish history in the side panel.

Figure 2



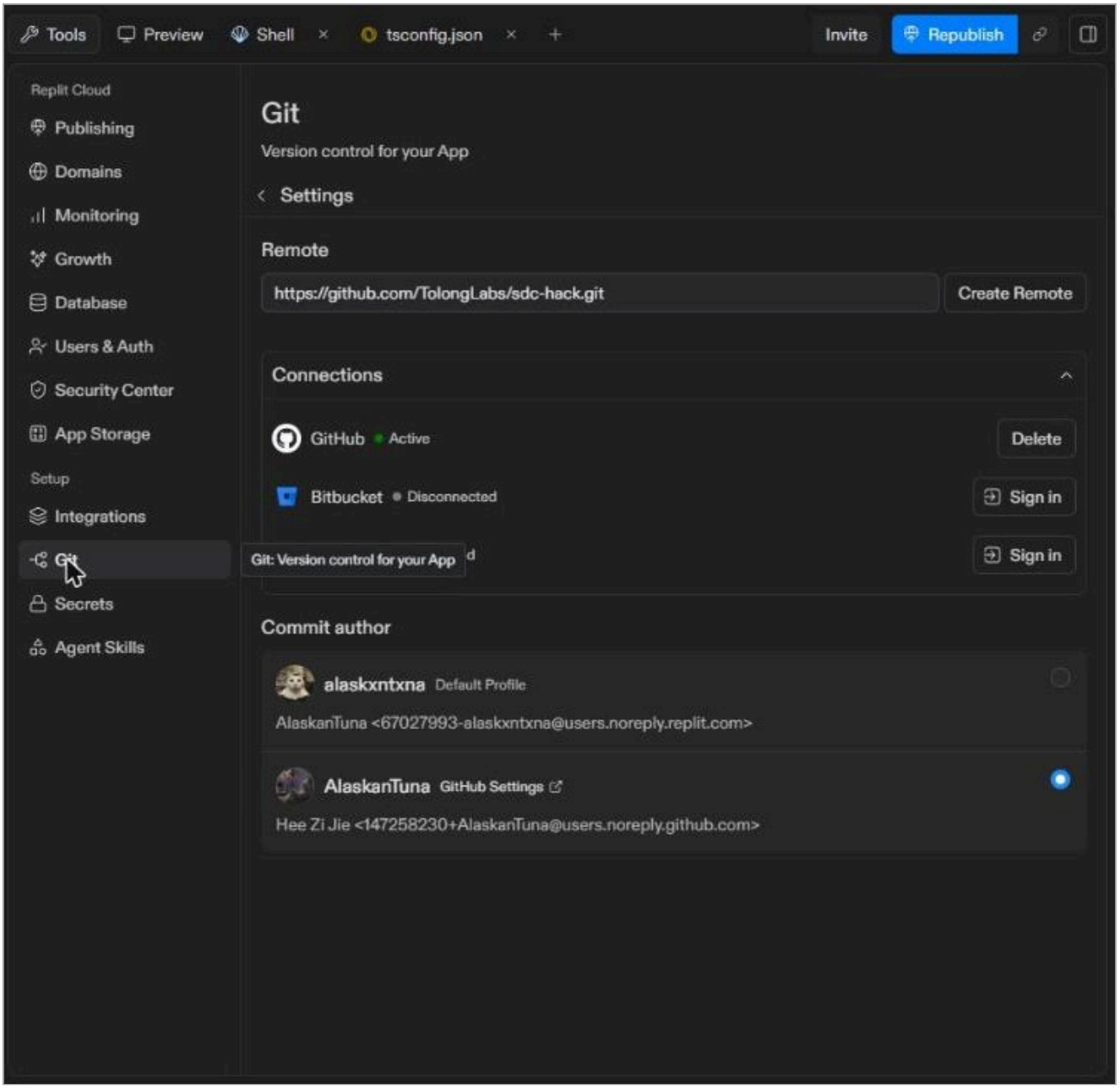
Publishing: the deployment is Live at kotak-alaskxntxna.replit.app, published by AlaskanTuna, public, on Autoscale in Asia.

Figure 3



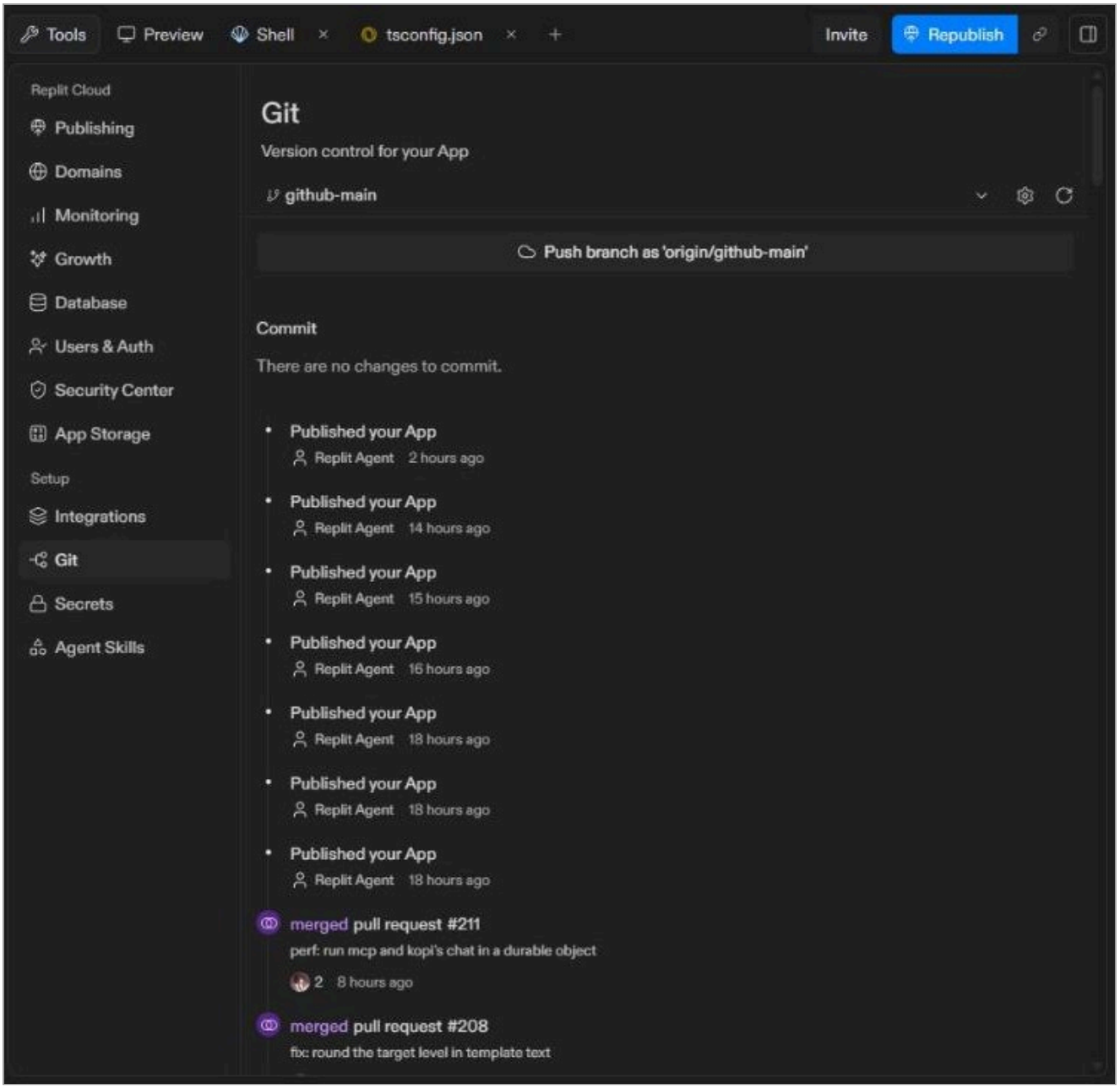
Publishing history: each publish carries its commit and the AlaskanTuna account.

Figure 4



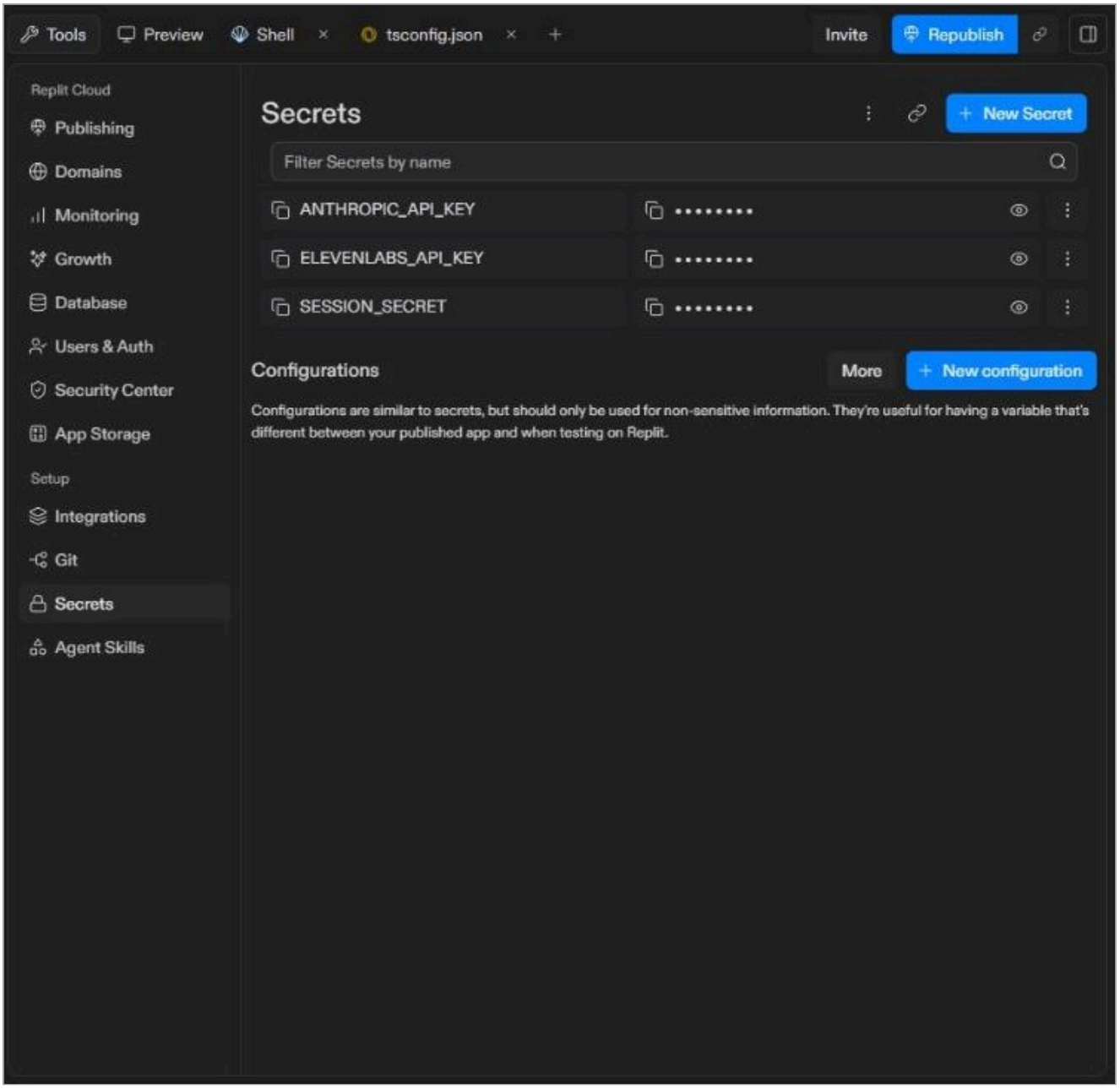
Git settings: the project's remote is the team's GitHub repository, and the GitHub connection is active.

Figure 5



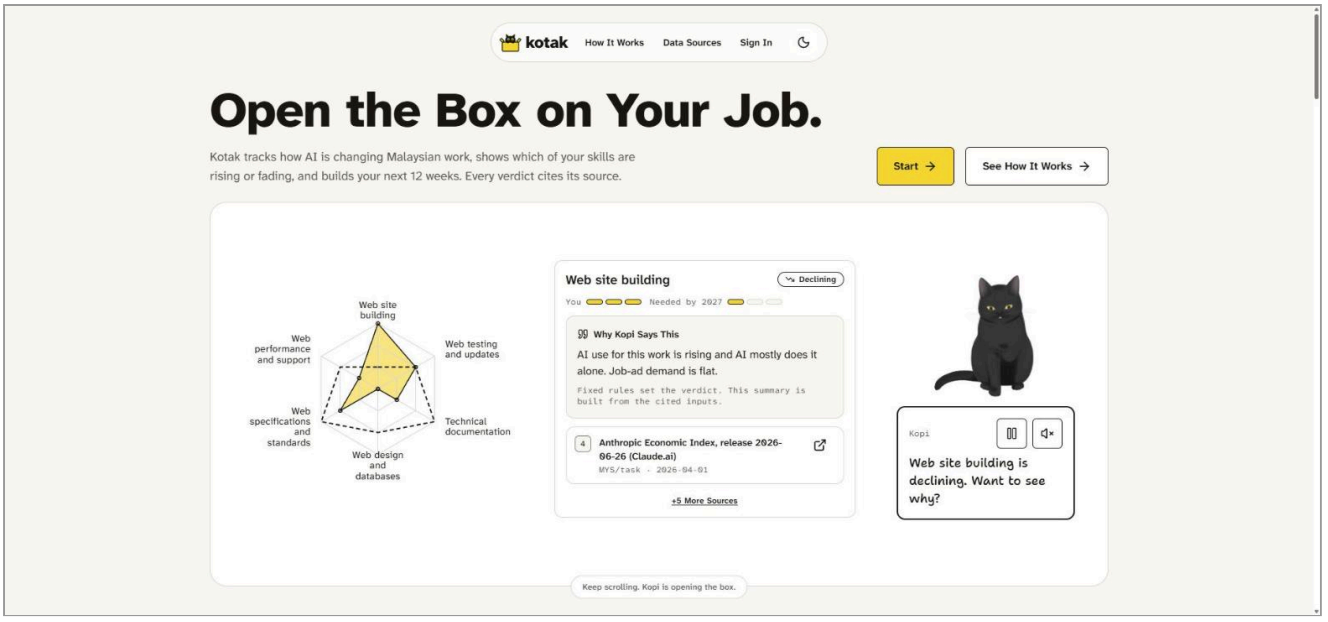
Git history on branch github-main: Replit publishes and the merged pull requests they include.

Figure 6



Secrets: API keys live in Replit Secrets, never in the code. Values stay hidden.

Figure 7



The published app, opened at https://kotak-alaskxntxna.replit.app on 23 October 2026.

Appendix C: RunPod Proof of Use

Handbook 10.7 asks for a RunPod dashboard screenshot showing the endpoint URL, the GPU type and the runtime, from the team's own account. The following pages show the serverless endpoint kotak-corners-batch (https://api.runpod.ai/v2/5yjk19uqmrpbp7) on one NVIDIA H100 80GB GPU, and the batch runtime. The same file is submission/RUNPOD PROOF OF USE.pdf in the repository.

RunPod Proof of Use

Kotak · SDC Hackathon 2026, Domain 1 · Batch run of 22 October 2026

What Ran on RunPod

RunPod account	Team account (AlaskanTuna), funded by the SDC RunPod voucher: US$25 credit redeemed 22 Oct 2026
Product	Serverless, queue-based endpoint `kotak-corners-batch`
Endpoint ID	`5yjk19uqmrpbp7`
Endpoint URL	`https://api.runpod.ai/v2/5yjk19uqmrpbp7` (OpenAI-compatible route `/openai/v1`)
Worker image	`runpod/worker-v1-vllm:v2.28.0` (vLLM v0.30.0, CUDA 13.0), the stock RunPod vLLM worker
Model	`Qwen/Qwen3.8-27B-FP8`, open weights under Apache 2.0
GPU	1 × NVIDIA H100 80GB HBM3 (H100 SXM), data centre AP-IN-1; pools 80 GB PRO and 80 GB; max 1 worker
Runtime	1285 s in total: smoke run 502 s (with cold start), full run 783 s
Volume	930 requests, 0 HTTP errors, 7 answers rejected by the validator and regenerated, 0 roles failed
Tokens	1,092,522 prompt, 1,137,450 completion
Cost	About US$1.25: account balance US$25.00 before, US$23.75 after (RunPod console, Billing page, 22 Oct 2026 16:05 MYT)

Workload

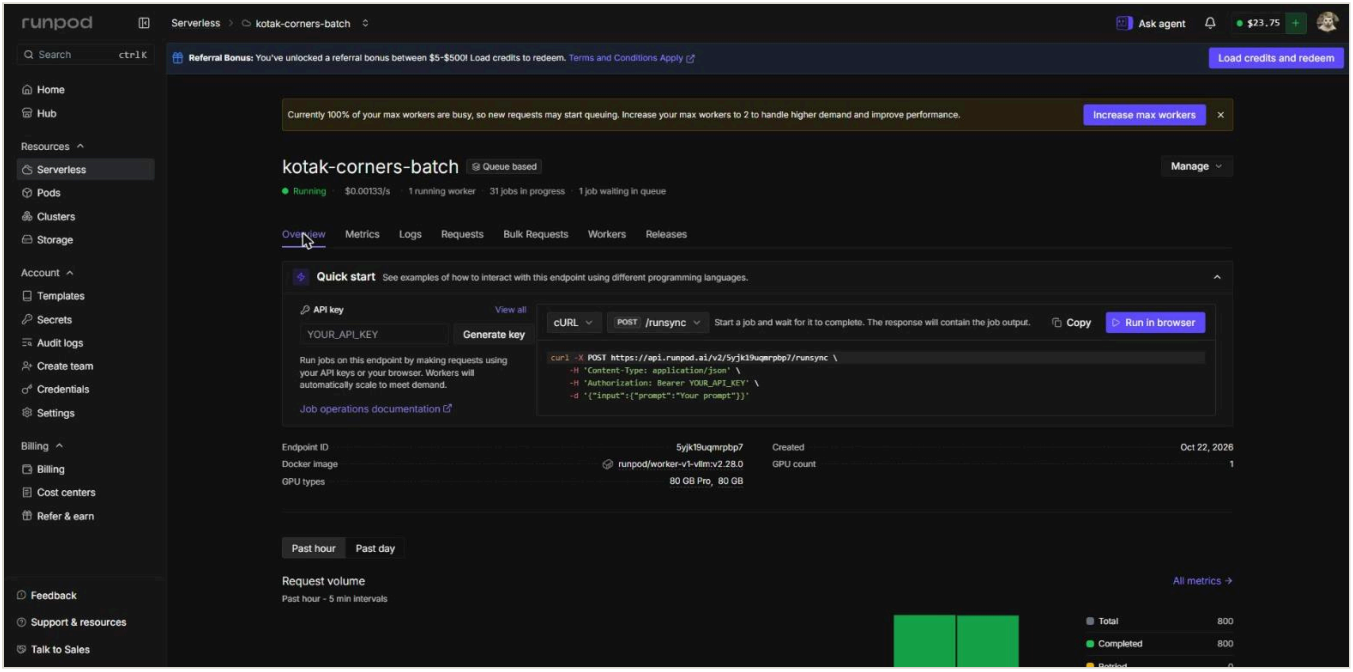
Kotak turns any job into six skill corners and asks twelve yes/no questions about them. RunPod generates that content once, ahead of time, for every one of the 923 O*NET occupations. For each occupation, `scripts/corners/run-batch.ts` sends the job's O*NET tasks to the model with a strict JSON schema. The model groups the tasks into six named corners and writes three can-do statements per corner (basic, working, advanced), with learning hours, SDG targets and search-link resources. A validator rejects any answer that drops a task, puts a task in two corners or cites a task the job does not have, and the runner retries it. The result, `data/generated/role-corners.json`, feeds the check-up questions, the Future Skills Radar and the roadmap; `role-corners-run.json` records every run.

There is no custom handler: the stock RunPod vLLM worker serves the model, and the endpoint settings are documented in `runpod/README.md`.

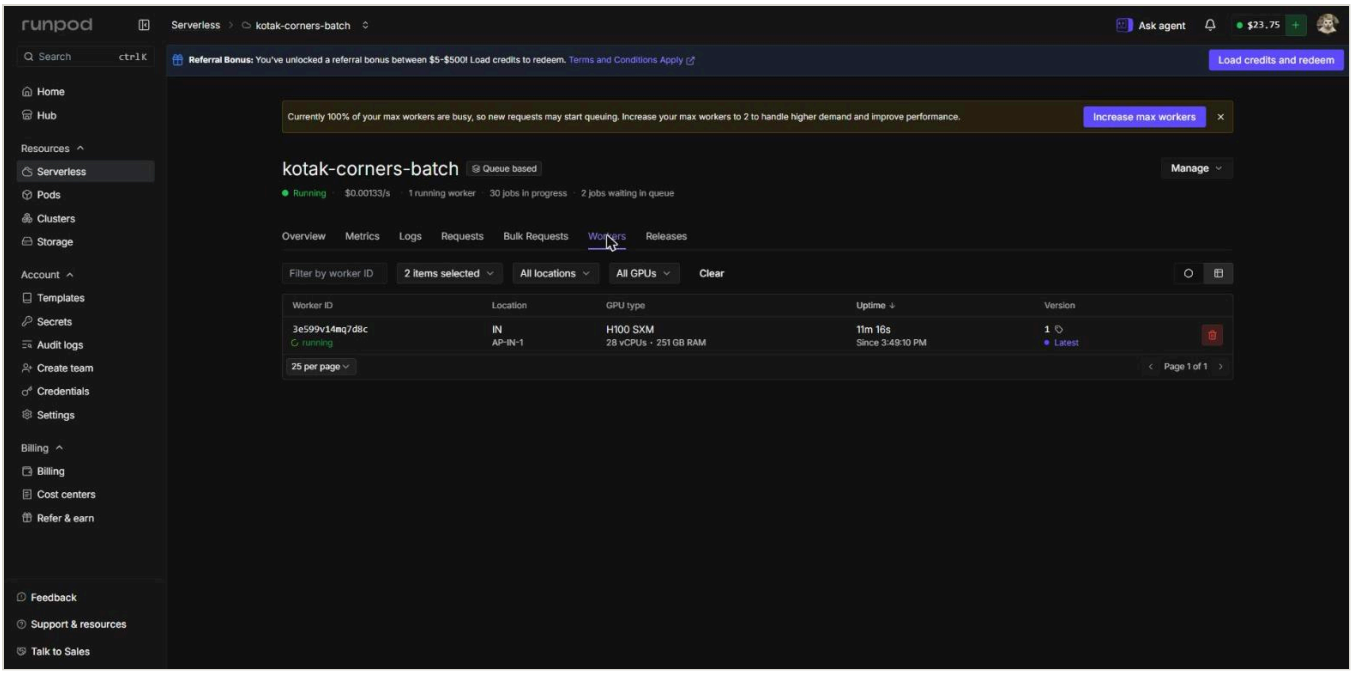
Example Request

```
curl https://api.runpod.ai/v2/5yjk19uqmrpbp7/openai/v1/chat/completions \
  -H "Authorization: Bearer $RUNPOD_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{
    "model": "Qwen/Qwen3.8-27B-FP8",
    "messages": [{"role": "user", "content": "Name one skill of a graphic designer as
JSON."}],
    "temperature": 0,
    "max_tokens": 50,
    "chat_template_kwargs": {"enable_thinking": false},
    "response_format": {"type": "json_schema", "json_schema": {"name": "skill", "schema": {
      "type": "object", "properties": {"skill": {"type": "string"}},
      "required": ["skill"], "additionalProperties": false}}}
  }'
```

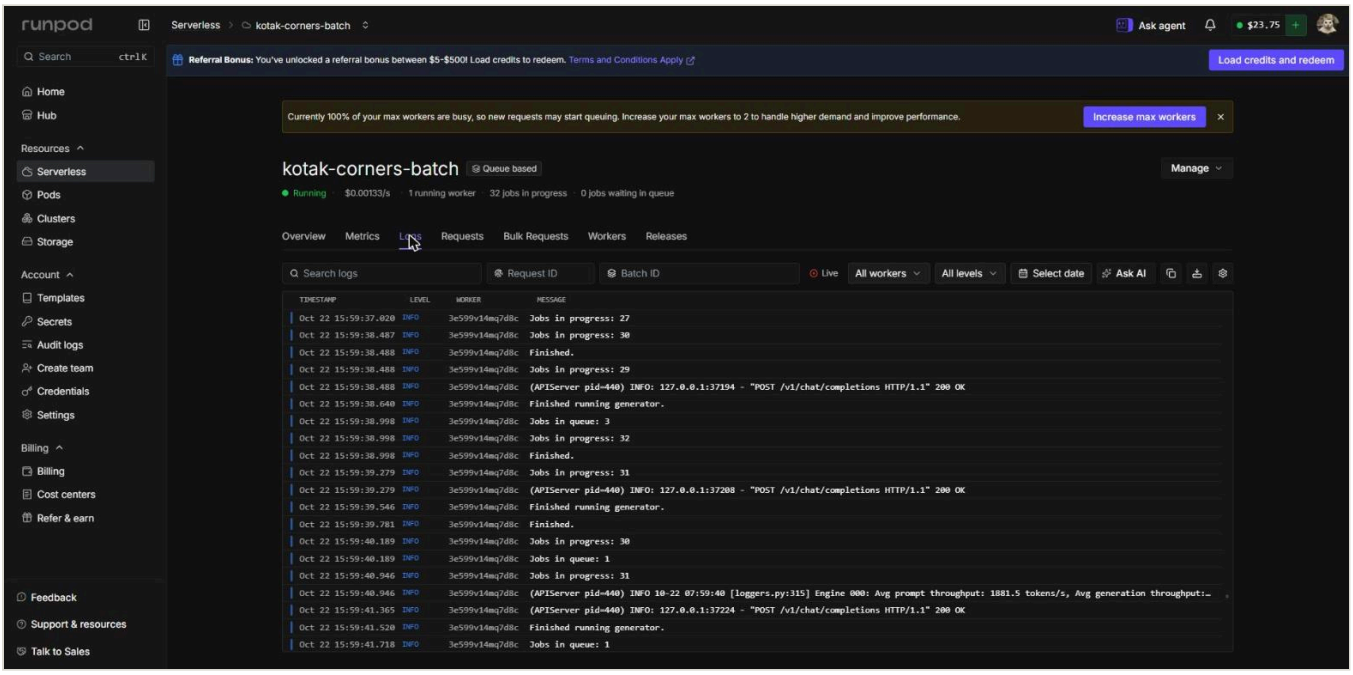
RunPod Console Screenshots



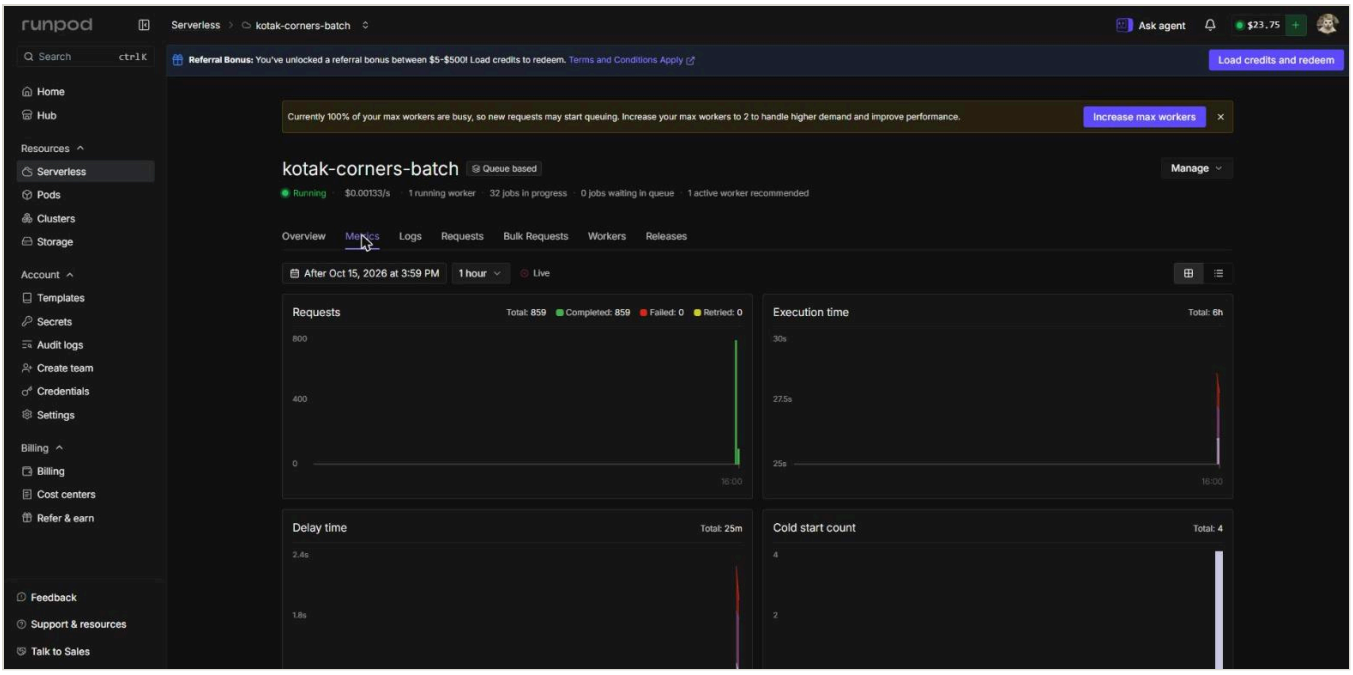
Endpoint overview: endpoint ID, worker image, GPU types, a cURL example and 800 completed requests, with the account balance top right. 22 Oct 2026, 16:00 MYT.



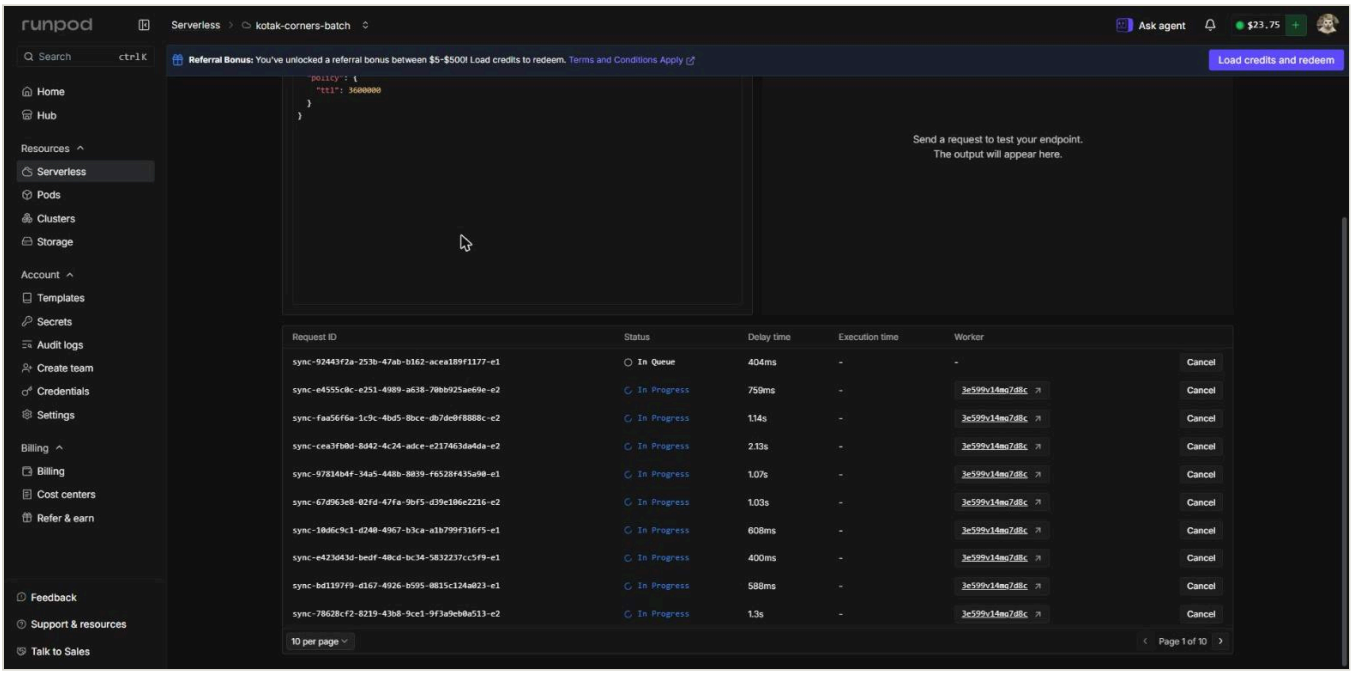
Workers tab during the full run: worker 3e599v14mq7d8c on an H100 SXM in AP-IN-1, running for 11 min.



Logs tab during the full run: vLLM serving POST /v1/chat/completions with 200 OK, about 30 jobs in progress.



Metrics tab: 859 requests completed, 0 failed, 0 retried at capture time; 4 cold starts across both runs.



Requests tab: batch jobs in progress on worker 3e599v14mq7d8c.